

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Energy conservation: transport, buildings, industry - Set 1

Mark scheme

AQA spec 3.3.4.2; Energy resources; textbook pp. 225-235

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• $(2.9 - 0.7) / 2.9 \times 100 = 75.9\%$ (accept 76%)	1
01.2	• Although the larger vehicle uses more energy in total, it carries many more people / a larger load • So the energy per passenger (or per tonne of goods) is lower; steel wheels on rails also have low rolling resistance	2
01.3	<i>Any 3 from:</i> • Flexibility of routes / destinations and door-to-door convenience • Time / speed and reliability where public transport is infrequent • Low fuel tax / cheap fuel • Carrying goods; comfort; status	3
01.4	• Integrated systems (park and ride, bicycles on trains, linked timetables) make it convenient to use the most efficient mode for each part of a journey • So more journeys shift from cars to bulk transport / cycling • ATM / smart motorways use variable speed limits to prevent congestion and keep traffic flowing freely • Vehicles avoid stop-start driving and run nearer their optimum speed (about 56 mph), reducing fuel wasted in acceleration and idling	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-228; AQA spec 3.3.4.2.1.	

Question	Marking guidance	Marks
02.1	• Aerodynamic shapes / fairings reduce friction with the air (air resistance) so less energy is needed to propel the vehicle, especially at speed; hydrodynamics for ships • Reduced mass (plastics, carbon-fibre composites, high-strength steel alloys, aluminium engine blocks, lighter batteries and magnets, rounded shapes) means less energy to accelerate and less rolling resistance, so lower fuel consumption • Conventional friction brakes convert kinetic energy to waste heat; regenerative braking uses a generator to convert it to electricity stored in a battery (or returned to the rail grid) which then propels the vehicle, so less fuel is needed to accelerate again	6
02.2	• Pneumatic tyres deform as they rotate; the continual change in shape and air movement inside generates frictional heat (more if under-inflated) • Steel wheels do not deform so rolling resistance is very low	2
02.3	Any 2 from: • More valves per cylinder to remove exhaust gases fully so they do not mix with fresh fuel • Better temperature control (thermostatic fan) • Electronic ignition control / regular servicing to keep spark timing correct • Turbochargers supplying more oxygen	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-227; AQA spec 3.3.4.2.1.	
03.1	• The energy used to extract raw materials, manufacture the components and assemble the vehicle (for a typical car about 1.5 years of its fuel use) • Reduced by using recycled materials (eg recycled aluminium needs 1/20 of the energy of new) • Lighter designs using less material; longer vehicle life spreading the embodied energy	3
03.2	Any 4 from: • Use of recyclable materials wherever possible • Easy identification of components and their composition (code stamps) • Easy dismantling and separation of components / materials • Reusable components for new vehicles • Compostable materials for parts that cannot be recycled	4
03.3	• At low speed a large proportion of the energy just runs the engine rather than moving the car • At high speed energy is wasted overcoming air resistance (which rises with the square of speed); optimum for many cars about 56 mph	2
03.4	• Automatic stop / start systems cutting the engine when stationary; smooth driving in higher gears avoiding sudden braking / acceleration	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 227-228; AQA spec 3.3.4.2.1.	
04.1	• Single: $5.8 \times 20 \times 12 = 1392 \text{ W}$ • Argon: $1.6 \times 20 \times 12 = 384 \text{ W}$ • Reduction = 1008 W (accept 1010)	3
04.2	• A wider gap of trapped gas reduces conduction because gas is a poor conductor • If the gap is too wide convection currents start in the gas, carrying heat across; the optimum depends on the gas and window size	2

Question	Marking guidance	Marks
04.3	• Low-e coating reflects long-wavelength infrared back into the room, reducing radiation losses • Argon / krypton / xenon have lower thermal conductivity than air so less conduction across the gap	2
04.4	• Panes must be held apart by small spacers which cannot be long, so the gap is only about 0.2 mm • Used where there is not room for thick gas-filled multiple glazing (eg historic buildings)	2
04.5	• Each extra layer saves less energy (diminishing returns) while cost rises, so beyond the optimum the cost of insulation exceeds the value of energy saved	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-231; AQA spec 3.3.4.2.2.	
05.1	• Orientation: south-facing windows (northern hemisphere) maximise passive solar gains; rooms needing warmth on the sunny side; fewer / smaller windows on the north side where losses are greatest; solar screens / brise soleil prevent summer overheating • Low SA:V (compact shapes, terraced / joined buildings, fewer storeys of small footprint) means less external surface per unit volume through which heat is lost • Earth sheltering: the ground is warmer than winter air and shields the building from wind, so conduction and wind-chill losses are reduced	6
05.2	• Cement manufacture (heating limestone) is very energy intensive so concrete has high embodied energy • Limecrete has about half the embodied energy of concrete; rammed earth, straw bales and lime mortar are lower still	2
05.3	• Warm stale air leaving and cold fresh air entering pass through the exchanger in opposite directions (counter-current) separated by thin conducting pipes / plates with a large surface area • Heat transfers from the outgoing to the incoming air so fresh air enters already warmed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-232; AQA spec 3.3.4.2.2.	
06.1	• Occupancy sensors detect infrared / movement and switch lights or appliances off when no one is in the room • Programmable thermostats set different temperatures for different rooms and times so unused rooms / periods are not heated	2
06.2	• LEDs convert a much larger proportion of electricity to light rather than heat • Washing machines with cold / low-temperature cycles and fast spins (less drying); dishwashers using less water so less heating; cookers with double / triple glazed doors; fridges with linear compressors; LED screens using half the power of plasma; germanium transistors	3
06.3	<i>Any 3 from:</i> • Turning off lights / appliances rather than standby • Turning thermostats and radiator valves down / off in unused rooms • Only heating the water needed (kettle) • Showering instead of bathing; eco-cycles • Choosing smaller vehicles / less travel	3

Question	Marking guidance	Marks
06.4	• Stored hot water loses heat continuously through the tank walls and pipes • Heating as needed (or just before) avoids these standing losses; if storage is needed the tank should be well insulated	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 232-233; AQA spec 3.3.4.2.2.	
07.1	• Conventional: $40 / 100 = 40\%$ • CHP: $(35 + 45) / 100 = 80\%$	2
07.2	• Extracting less energy as electricity leaves the cooling / condenser water at a higher temperature • Hotter water is more useful for district heating of homes, fish farms, greenhouses	2
07.3	• Hot water loses heat and needs pumping energy over distance; heat cannot be transported as far as electricity	1
07.4	• Long narrow pipes give a large surface area and copper is a good thermal conductor so heat transfers quickly • Hot and cold fluids flowing in opposite directions maintain a temperature gradient along the whole length so more heat is transferred • Slower flow allows more time for heat to transfer	3
07.5	• One large tank has a lower surface area : volume ratio than many small ones • A sphere has the smallest surface area for any volume; less surface means less heat loss (with insulation of low thermal conductivity)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 233-235; AQA spec 3.3.4.2.3.	
08.1	• Molten iron from the blast furnace is converted directly to steel on the same site • So it does not cool and need re-melting; transport energy between sites is also avoided; waste heat from one process used by another (eg Kalundborg)	2
08.2	• Collecting and processing small dispersed quantities can use more energy in transport than is saved • Mass production and large-scale mining have economies of scale; some materials (glass) take more energy to make than plastic but can be refilled many times	2
08.3	• Transmitting at high voltage means low current so less energy is lost as heat in the cables (loss proportional to current squared); overhead cables cheaper than underground which need cooling • ICT / smart grids co-ordinate real-time data on demand and supply from many sources, switching generation, storage (pumped storage, batteries) and flexible loads so surplus is not wasted and inefficient peaking plants are avoided	4
08.4	<i>Any 2 from:</i> • Proximity to the grid / demand to reduce transmission losses and cable costs • Access to cooling water for thermal stations • Resource availability (wind, tidal range, fuel deliveries) • Land availability and planning / public acceptance; stable geology	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 234-235; AQA spec 3.3.4.2.3.	